

ASAD AMAD SHEIKH

Backend Engineer · Python · FastAPI · PostgreSQL · Distributed Data Systems

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SUMMARY

Backend engineer working on data-intensive services in Python and FastAPI. I own systems end to end — schema design, API, deployment, and the infrastructure underneath — and currently run a live product I built and shipped alone. Prior work in government fleet telemetry and production telecom data.

EXPERIENCE

Founder & Backend Engineer | ParchiVisa

2025 – Present

FastAPI · PostgreSQL · Redis · Next.js · DigitalOcean · Cloudflare R2 · Clerk · Playwright | parchivisa.app

- Sole engineer on a live visa-assessment product: FastAPI service, Postgres schema, Next.js frontend, and the infrastructure it runs on.
- Built four country-specific eligibility engines (UK, Australia, Canada, USA). Replaced proportional scoring with tiered band-capping after finding that proportional weights let a strong financial profile mask a disqualifying academic gap — the model now fails a candidate on any single hard blocker regardless of aggregate score.
- Report pipeline runs retrieval over a structured corpus of official immigration requirements into LLM narrative synthesis, with one React template rendered to both web and PDF via headless Playwright so the two outputs cannot diverge.
- Deployed and operate the service on DigitalOcean behind Cloudflare, with secrets injected at runtime and nothing credential-bearing in source control.
- Live end to end in production: intake, four-engine scoring, generated PDF report, and payment on a single deploy with no manual steps.

Backend Engineer | Frontier Works Organization

Jun – Aug 2025

Python · PostgreSQL · REST APIs · IoT telemetry · Flotilla fleet platform

- Built the data exchange layer between PostgreSQL and the Flotilla fleet platform, moving real-time telemetry across government fleet systems covering a fleet of over 300 vehicles.
- Replaced manual batch reporting with automated aggregation pipelines, cutting reporting latency from **30 mins** to **< 1 min**.
- Traced production timeout failures under peak concurrent load to unbounded queries and connection pool exhaustion; rewrote the queries and reconfigured pooling, eliminating the timeouts at the same peak load.

Data Engineer | Ufone (PTCL Group)

Jun – Aug 2024

SQL · Relational schema design · Production telecom data

- Optimized relational schemas across production telecom infrastructure handling high-volume SME datasets, covering **80 tables**.
- Found and fixed schema inefficiencies causing silent data loss in multi-departmental compliance pipelines, across a compliance dataset of roughly **670 GB**.

PROJECTS

VehicleWatch | Fleet Telemetry & Predictive Maintenance

Live

FastAPI · PostgreSQL · Redis · scikit-learn · Docker · Render | vehicle-watch.onrender.com/docs

- Ingests vehicle sensor readings every 2 seconds and flags mechanical faults before they escalate into breakdowns.
- Uses an Isolation Forest and Local Outlier Factor ensemble over a 10-dimensional feature space — chosen because fleet fault data arrives unlabelled, ruling out supervised classification — and emits a plain-English diagnostic per alert.
- Trained models persist in Redis under a dual-pool configuration, so scoring stays warm across restarts without retraining.
- Holds **sub-100ms** scoring latency end to end on a single container instance.

Veriloom | Skills-First Hiring Platform

Express.js · Prisma · PostgreSQL · Next.js · Clerk · Cloudflare R2 · LLM API | github.com/asadsheikh91/VeriLoom

- Two-sided hiring platform that replaces CV screening with verified work: organizations post task-based roles, candidates submit real deliverables, and an LLM evaluation pass scores the submission before any human review.

- Scores on two independent signals — deliverable quality, and comprehension measured in an AI-conducted follow-up interview keyed to the candidate's own submission — so work that was outsourced fails the second gate even when it passes the first.
- Submissions are isolated per organization on Cloudflare R2 behind Clerk auth, with a Prisma schema over PostgreSQL fronted by an Express API.

InfluencePay | ERC-20 Escrow Marketplace for Influencer–Brand Deals / github.com/asadsheikh91/influence *Personal Project*
Solidity 0.8.24 · Hardhat · OpenZeppelin · ethers.js v6 · Ethereum Sepolia · Next.js 14 · Express · Prisma · PostgreSQL · Clerk

- Escrow marketplace settling brand–creator deals through Solidity smart contracts rather than a custodial account, so neither party holds the other's funds while a deliverable is outstanding.
- Issued a custom ERC-20 settlement token on OpenZeppelin's audited base, with a tier-based fee schedule applied on release so platform take varies by creator tier without a separate off-chain billing path.
- Modeled escrow as an on-chain state machine — fund, lock against milestone, release on acceptance, explicit dispute path — deployed to Ethereum Sepolia via Hardhat and tested through the full lifecycle with ethers.js v6.

TECHNICAL SKILLS

Languages Python, JavaScript / TypeScript, SQL, Java

Backend FastAPI, Express.js, SQLAlchemy, Pydantic v2, Prisma, asyncio, background workers, REST / OpenAPI design

Data PostgreSQL, Redis (caching, dual-pool, model persistence), Alembic, schema design, query optimization

AI / ML LLM integration (Claude, Gemini), retrieval over structured corpora, structured output, scikit-learn, Isolation Forest, Local Outlier Factor

Infrastructure Docker, GitHub Actions, Nginx, Gunicorn, DigitalOcean, AWS (EC2 / RDS / S3), Cloudflare (R2, DNS), Render, Linux

Security JWT, RBAC, Clerk, rate limiting, runtime secret injection, structured audit logging

Practices Schema migrations, CI/CD pipelines, containerised deploys, structured logging

Testing pytest (unit, async, coverage), httpx integration tests, schemathesis contract tests against OpenAPI, Locust load testing, ruff · mypy in C

EDUCATION

B.S. Computer Science | FAST NUCES, Islamabad
Graduated June 2026.

2022 – 2026

ACHIEVEMENTS

- Certificate of Appreciation, Frontier Works Organization (Aug 2025) — Sole recipient among a cohort of 45 interns.
- Selected, League of Launchers S2 — National Incubation Centre Islamabad (ParchiVisa).
- FastAPI & Backend Development — Packt Publishing.